

SMART INDIA HACKATHON 2025

Problem Statement ID – **SIH25002**

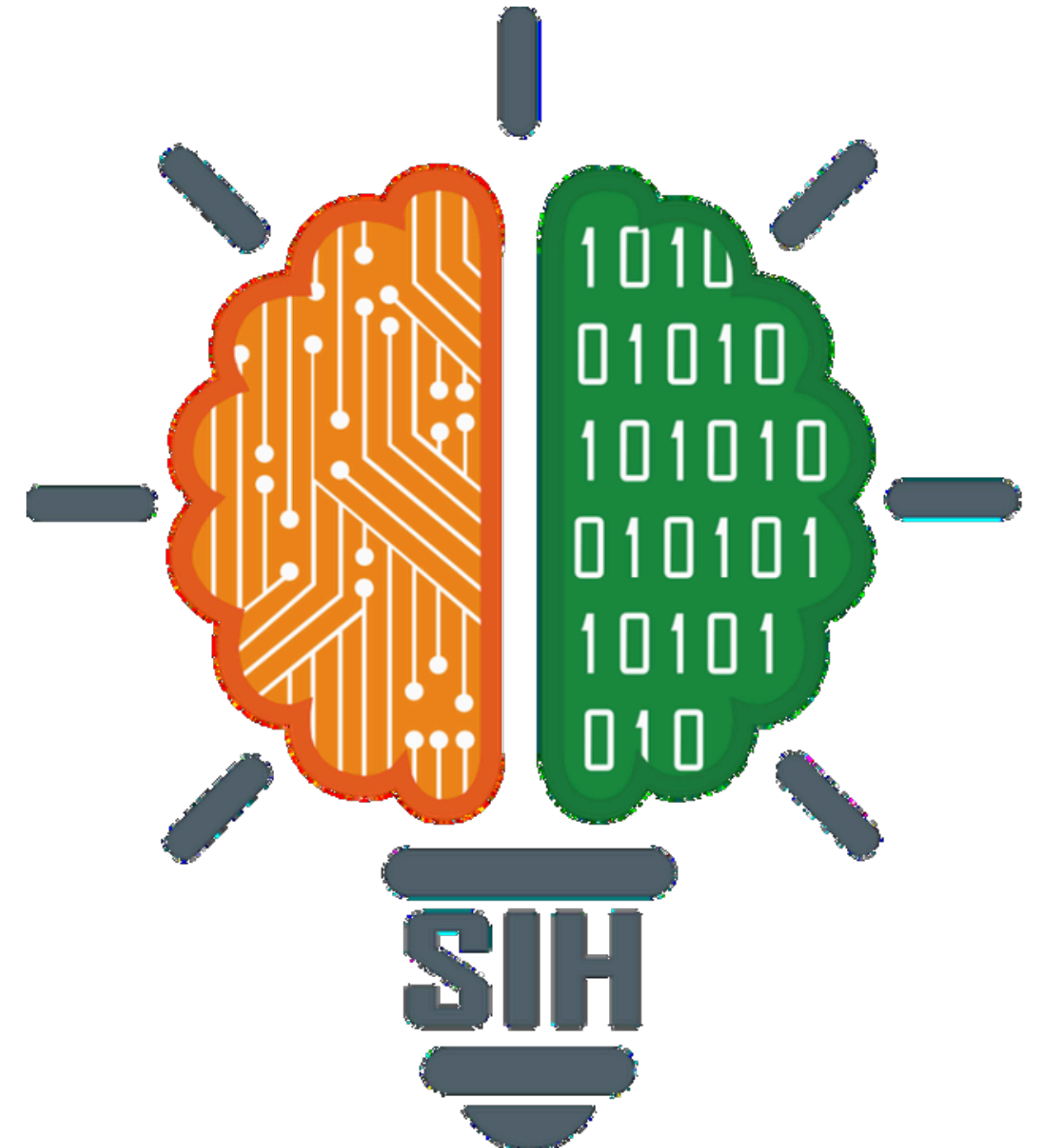
Problem Statement Title – **Smart Tourist Safety Monitoring & Incident Response System using AI, Geo-Fencing, and Blockchain-based Digital ID**

Theme – **Travel & Tourism**

PS Category – **Software**

Organisation – **Ministry of Development of North Eastern Region (MDoNER)**

Team Name – **STS Safety**





Understanding The Problem

- **Delayed Detection:** No live geo-fence means restricted-zone, missing-person and signal-loss events are identified **too late**.
- **Limited Context:** Generic tourist apps do not capture **North-East terrain**, Inner Line, forest cores or checkpoint reality.
- **Fragmented Insights:** Check-post, hotel, forest office and police **do not share a database** — a Tawang desk cannot verify a Guwahati ID.
- **Ineffective Support:** SOS that depends on a cloud round-trip **fails when the hills go dark**.
- **Poor Autonomy:** 7 of 8 NE states have **no dedicated Tourist Police**; NCRB has no domestic-tourist crime series.

Our Unique Solutions

- **Safety path is the database.** Blockchain and AI sit **beside it**. Unplug WAN: Realtime + Anvil + ONNX still demo.
- **Lawful identity, not an NFT.** Hotel learns match / no-match, **never the passport number**.
- **Evidentiary E-FIR.** Officer must approve. PDF hash is anchored. LLM **never files to CCTNS**.
- **Honest ML.** Synthetic NE tracks (DPDP). Hold-out: **0 FP on 1,500** normal-trek windows.
- **Zero-cost infrastructure.** Vercel Hobby + Supabase Free + HF CPU + Amoy. **No credit card** in the stack.
- **Five locales.** en · hi · as · bn · ne matching the PS multilingual clause.
- **Measured SOS.** **180–450 ms** first channel on the local stack; Telegram target < 2 s.

Our Solution

- **Real-Time Geo-fence:** PostGIS AFTER INSERT trigger — the phone **cannot skip an incident**.
- **Soulbound Digital ID:** ERC-5192 TDID; chain stores only keccak256(KYC || salt), **never PII**.
- **Hold-to-SOS:** 1.5 s hold writes a critical incident under RLS; **SMS fallback** if the insert fails.
- **On-device Warning:** Turf.js + IndexedDB + Background Sync for Arunachal towers.
- **AI beside the path:** IsolationForest ranks; Groq/Gemini draft E-FIR. **Alert still fires** if they die.
- **Live Command Room:** Critical-first queue, nearest-unit dispatch, MTTA/MTTR on **one map**.
- **DPDP by design:** Consent, pgcrypto KYC, RLS, 24 h ping purge, emergency contacts only at critical.

USPs {

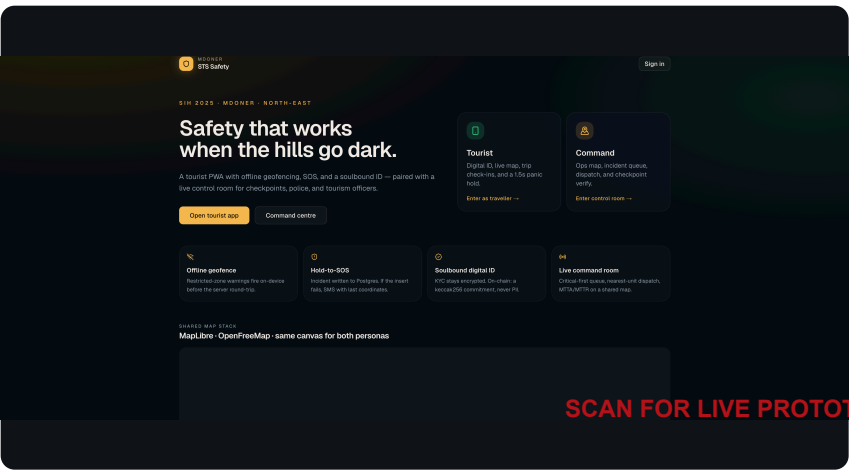
Real-time geo-fence for instant restricted-zone checks.

Lightweight stack tuned for hill connectivity.

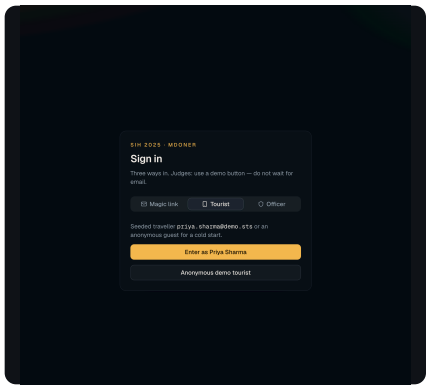
On-device micro-warnings to keep tourists safe.

Seamless command-room dispatch, even offline.

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SCAN FOR LIVE PROTOTYPE <https://sts-safety.vercel.app>

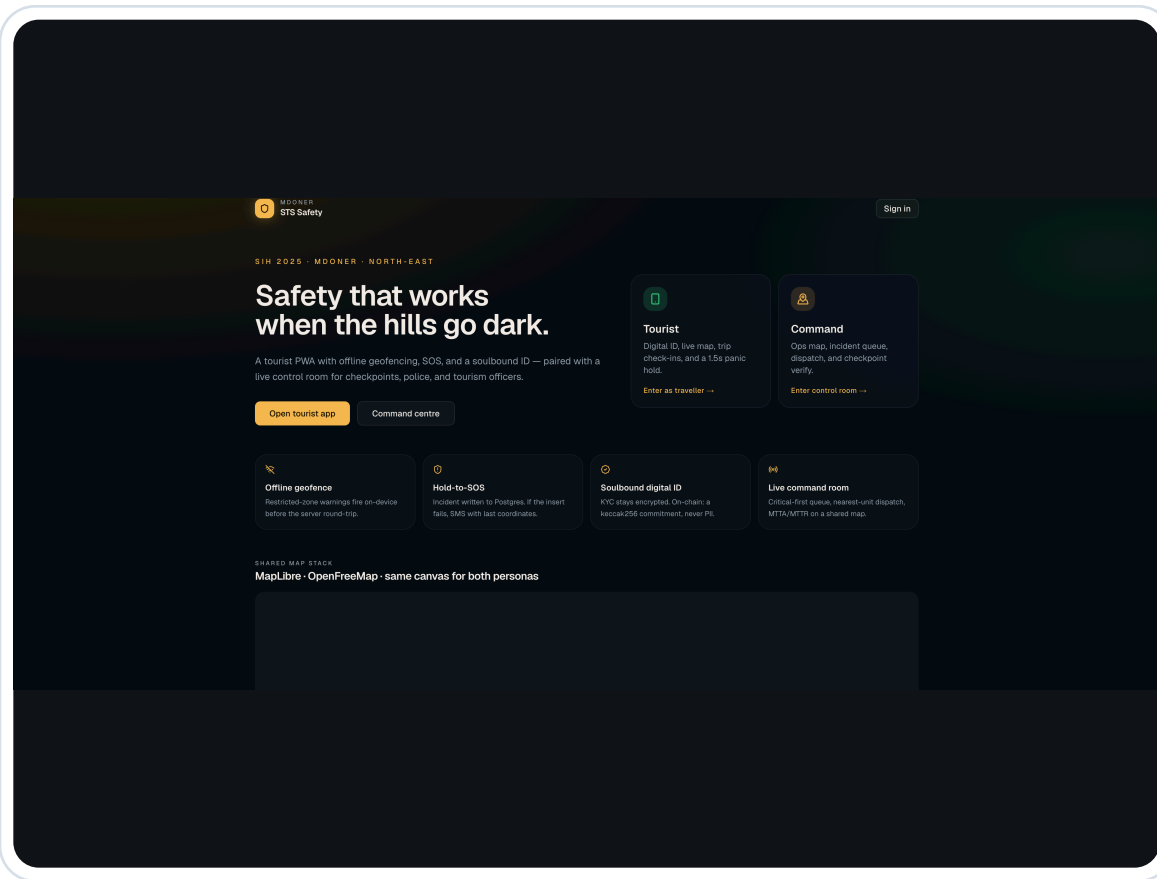


Technical Approach

Methodology

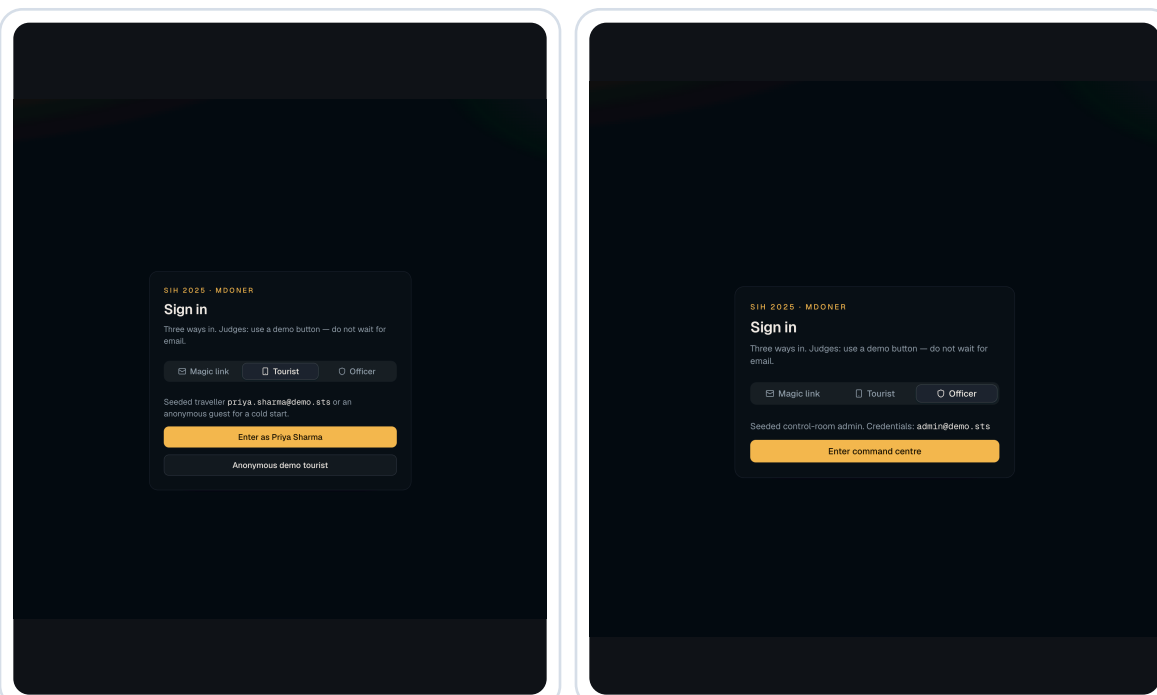
- **Multimodal Data Acquisition:** Capture tourist **GPS** (watchPosition), zone polygons, itinerary LineString, **SOS gesture** and optional IoT-band pings from the simulator.
- **On-device Pre-check:** **Turf.js** evaluates cached GeoJSON zones, vibrates a warning, queues pings in **IndexedDB** if the tower is gone.
- **Hot-path Ingest:** **RLS insert** into location_pings (~80–150 ms). No serverless hop. AFTER INSERT → **evaluate_position()**.
- **Geo Feature Extraction:** **ST_Covers** / **ST_DWithin**, corridor deviation, implausible speed, dwell, silence (pg_cron 20 min).
- **Fusion & Decision:** **Rules always create** the incident. IsolationForest **only ranks**. Groq → Gemini → template writes the brief.
- **Dispatch & Evidence:** Nearest on-duty unit, Telegram + Web Push + email. High/SOS records **hash-anchored** on Polygon Amoy / Anvil.

User Interface



Tourist PWA landing · dual persona entry

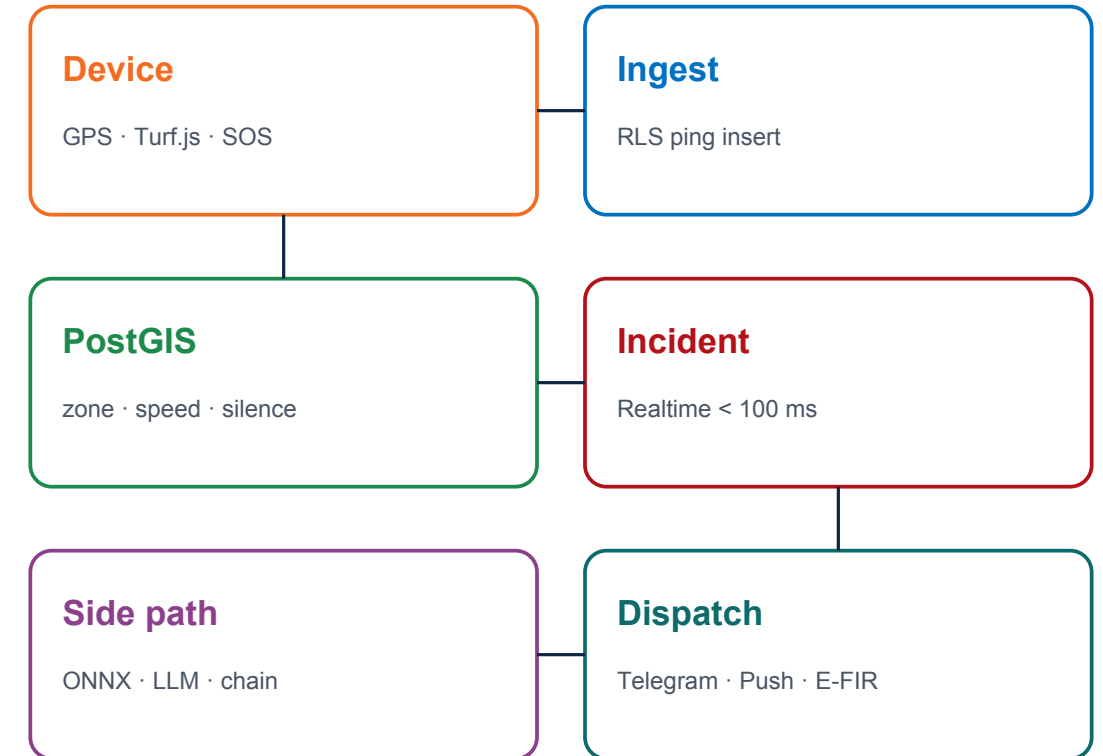
Admin / Command



Tourist demo login

Officer / command login

Implementation Flow



Tech Stack

Frontend

Next.js 16 PWA · MapLibre · Tailwind + shadcn · Turf.js

Geo & Backend

PostGIS 3 · Supabase RLS · PostgREST · pg_cron / pg_net

AI & Chain

ONNX IForest · Groq / Gemini · Polygon Amoy · ERC-5192 TDID

Panic path — nothing optional between button and siren

Hold 1.5 s → INSERT incidents {type:sos, severity:critical} under RLS (no Route Handler) → trigger skips debounce → Realtime full-screen takeover → Telegram + Web Push + email in parallel → 2 s ping cadence for 30 min.

Offline: Background Sync retries the insert; PWA opens a pre-composed sms: URI with last coordinates.

Design rule: blockchain and AI are enhancements on the side of the safety path, never inside it.

Surfaces: /home /sos /onboard /dashboard /verify
Simulator stands in for the IoT band on the same ingest path.

IMPACT AND BENEFITS



Potential Impact on Tourists

- **Early Danger Detection** – Restricted-zone, signal-loss and SOS signs fire **before** a missing-person file.
- **On-device Stability** – Turf.js warnings and hold-to-SOS work **when the tower is gone**.
- **Companion ID** – Soulbound QR at hotel / check-post / forest office — green, amber or red in **one scan**.
- **Task Readiness** – Control room sees **last-known point**, digital ID and nearest unit together.
- **Adaptive Support** – **Five UI languages** (en, hi, as, bn, ne) matching the PS multilingual clause.

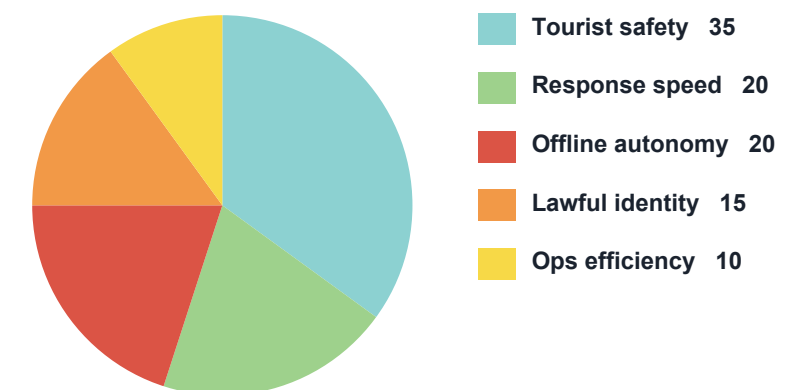
Mission & Operational Benefit

- **Risk Reduction** – **Continuous geo-fence** lowers illegal entry into Kaziranga core and border approaches.
- **Autonomous Safety** – **Works offline**: IndexedDB, Anvil, ONNX, PMTiles. Reliability during blackouts.
- **Efficient Crew Management** – Nearest on-duty dispatch; **MTTA / MTTR** become measurable.
- **Extended Coverage** – Force-multiplier for 7 NE states with **no Tourist Police** unit.
- **Crisis Handling** – Escalates SOS **instantly** with logs, chain proof and an officer-approved E-FIR draft.

Economic and Strategic Gains

- **Tourism is national-scale** – **5.22% of GDP**, 8.46 crore jobs, 9.95 million FTA (2024). NE still takes only 2.05% of FTAs.
- **Perception of safety** – A **binding constraint** on high-value / foreign itineraries into the hills.
- **Rs.0 to run the prototype** - First paid production line is SMS + a non-pausing database, not a rewrite.
- **Paperwork cut** – E-FIR drafts cut officer time; the **officer still signs**. Nothing is filed to CCTNS automatically.

Proportional impact across key domains.



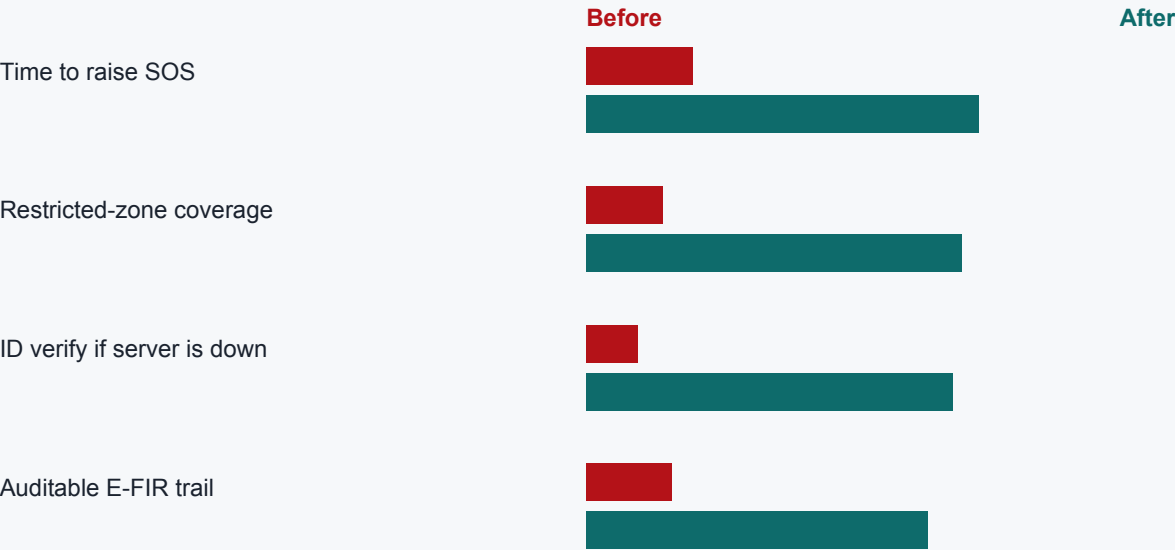
FEASIBILITY AND VIABILITY



Analysis of the feasibility of the idea:

- **Proven research base:** PostGIS containment and IsolationForest are validated; metrics are **published in-repo**.
- **Hardware compatibility:** **PWA + edge ONNX** run on a phone and a laptop. No custom spacecraft hardware.
- **Scalability:** **Same geofence engine** from 50 demo tourists to a district control room.
- **Training data:** Synthetic NE trajectories (**DPDP**). Rules catch what the forest misses.
- **Operational reliability:** Four env switches: DB / chain / AI / tiles. **Rehearsed unplugged**.
- **Energy / cost:** **Rs.0 infra**. IsolationForest hold-out P=1.00 R=0.56 F1=0.72 · 0 FP / 1,500.

Control-room capability - Before vs After



Bars are qualitative capability scores (0–100), not invented crime-drop %.

WHAT IFs....?

01

AI mis-ranks a trek?

Rules + PostGIS still decide. IsolationForest only escalates. rules-only is one env flip.

02

hardware / WAN dies mid-demo?

IndexedDB + sms: SOS. Anvil + ONNX + PMTiles. Four local fallbacks.

03

tourists reject tracking?

Opt-in, trip-scoped, purpose = safety. Visible score on /home. Not a camera grid.

04

extreme GPS / spoofed phone?

Refuse >150 km/h. Session JWT. SOS is a user gesture. Checkpoint QR binds the person.

05

LLM drafts a bad E-FIR?

Draft PDF only. approved_by is a human. Nothing is submitted to CCTNS.

Strategies to Overcome Challenges

Limited connectivity (Arunachal / hills)

Strategy: IndexedDB queue + Background Sync. Turf.js warns on-device. SOS falls back to sms: with coordinates.

Accuracy with limited real GPS (DPDP)

Strategy: Stated in the pitch. Decision to alert is rules + PostGIS. Forest is unsupervised on generated NE tracks.

Trust & engagement / “is this surveillance?”

Strategy: Consent primer, purpose limitation, RLS on every table, 24 h ping retention, emergency contacts only at critical

RESEARCH & REFERENCES



Feature	Our Proposed Solution	Existing Conventional Solutions
Geo-fence	PostGIS trigger; client cannot skip	App if-else or Mongo \$geoWithin
Digital ID	ERC-5192 soulbound + keccak256 commitment	QR in a database, or NFT with PII
AI role	Ranks only. Rules decide. LLM never gates SOS	LLM on GPS / “predictive hotspot”
Offline	Turf.js + IndexedDB + sms: + Anvil + ONNX	Needs internet / paid Map SDK
Privacy	RLS, pgcrypto, 24 h ping purge, no PII on-chain	Central Aadhaar log
E-FIR	Officer-approved PDF; hash anchored	Chatbot “files” a legal document
Cost	Rs.0, no card, four local fallbacks	Twilio + Mapbox + Firebase Blaze
ML honesty	Synthetic tracks; metrics published	Unstated accuracy on unseen GPS

Research and Datasets references

- [MoT / PIB — Tourism in NER, DTV+FTV 2023](https://tourism.gov.in/sites/default/files/2024-12/PIB2082598.pdf)
https://tourism.gov.in/sites/default/files/2024-12/PIB2082598.pdf
- [MoT Compendium Key Highlights 2024 \(NE FTA 2.05%\)](https://tourism.gov.in/sites/default/files/2025-02/India%20Tourism%20Data%20Co...)
https://tourism.gov.in/sites/default/files/2025-02/India%20Tourism%20Data%20Co...
- [PIB — FTA 9.95M \(2024\); GDP 5.22%; jobs 84.63M](https://www.pib.gov.in/PressReleasePage.aspx?PRID=2240657)
https://www.pib.gov.in/PressReleasePage.aspx?PRID=2240657
- [Lok Sabha USQ 1299 \(27.07.2026\) — Tourist Police + NCRB 156](https://sansad.in/getFile/lsapps/loksabhaquestions/annex/188/AU1299_510Xy0.pdf)
https://sansad.in/getFile/lsapps/loksabhaquestions/annex/188/AU1299_510Xy0.pdf
- DPDP Act 2023 · ERC-5192 · W3C Verifiable Credentials
- Liu et al., Isolation Forest, ICDM 2008
- PostGIS ST_Covers / GiST spatial index
- SIH PS SIH25002 — MDoNER, Travel & Tourism, Software

Future scope — only what the architecture already allows

- Ministry SMS short-code on the existing notification adapter.
- Real Inner Line Permit / KYC adapter; custodial HD wallet → ERC-4337 paymaster.
- Polygon mainnet + hardware-backed issuer. Contracts are vanilla OZ v5 — config, not a rewrite.
- Physical IoT band posting to the same location_pings ingest (the simulator is that interface today).
- CERT-In intake + threat model. Geofence engine stays in Postgres.

Prototype: tourist PWA + command centre + checkpoint verify + PostGIS engine + ONNX model + Anvil/Amoy contracts.
Demo: docs/DEMO-SCRIPT.md Hostile Q&A: docs/JUDGE-QA.md